

Exact limits of Krylov-complexity comparisons under purification

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Abstract

Purification relates mixed quantum states to pure states on a larger Hilbert space, but it changes the spectral weights that determine Krylov complexity. We examine the comparison conjectures of Das and Mori using their normalized Hilbert–Schmidt and spread-complexity conventions. Each side of their two-sided hierarchy admits an exact full-rank qutrit counterexample, with separate rational finite-time certificates. The hierarchy holds for every qubit, so physical dimension three is minimal for failure of either side. At fixed dimension four and fixed Hamiltonian, two different full-rank families have unbounded recurrence-period coefficients of variation for the purification-to-mixed complexity ratio and its reciprocal, respectively. A third family has fixed purity and an unbounded ratio and temporal mean. These are no-go theorems for explicit universal concentration and purity-only control formulations motivated by the qualitative second conjecture. In contrast, the pure-density factor-two inequality holds in finite dimension; we give a self-contained proof and identify it as a corollary of existing tensor-product superadditivity. A qubit concentration bound and state-dependent return bounds delimit the surviving comparisons. The common mechanism is the distinct weighting of coherent sectors by the density matrix and its positive square root.

1 Introduction and source problem

Krylov constructions convert unitary evolution into motion along a basis generated by successive applications of the dynamical generator. Operator complexity measures the mean position in an operator Krylov chain, while spread complexity uses a state-vector chain [5, 1]. A density matrix can participate in either construction: it is itself an operator, and it also specifies a pure state after purification. Comparing these descriptions requires control of both the seed normalization and the generator. Earlier work on density-matrix operators already distinguishes their Krylov complexity from the spread complexity of a pure state [2].

Das and Mori [3] formulated this comparison for two unitary purification schemes. The time-independent scheme evolves only the physical system, while the time-dependent scheme evolves the ancilla by the conjugate unitary. Their Eq. (4) proposes

$$\mathcal{C}_S(\Psi_\rho^{U*}(t)) \leq \mathcal{C}_K(\rho(t)) \leq \mathcal{C}_K(\Psi_\rho^I(t)). \quad (1.1)$$

Here the rightmost quantity is the operator complexity of the rank-one density matrix of the purification. It is not the spread complexity of the purified vector. We retain the published source’s conventions throughout.

The source also proposes two different kinds of statement. Its second conjecture describes the ratio of the time-dependent purification’s spread complexity to the mixed-state operator complexity as approximately constant relative to its temporal mean, with a value depending on purity. This is qualitative: it specifies neither a uniform tolerance nor an exact function of purity. The third conjecture is the precise factor-two inequality between operator and spread

complexities of the same purification. The distinction between these statements is central to the analysis below.

We give separate exact qutrit witnesses against the two sides of Eq. (1.1). The source's qubit hierarchy remains valid; its explicit three-atom reduction proves that neither inequality can fail in a smaller physical dimension. For temporal ratios, we state and rule out three universal formulations: a finite uniform recurrence-period coefficient of variation for the main ratio, the corresponding bound for its reciprocal, and finite purity-only upper control of the main ratio or its mean. The two concentration questions are logically distinct, so they require different families. All counterexamples use strictly positive density matrices at every allowed parameter value.

The factor-two inequality has a different status. Tensor-product superadditivity, stated by Murugan and van Zyl [4, Sec. 2, Eq. (2.4)], implies it after vectorizing a rank-one projector. We give the needed finite-dimensional proof in terms of nested polynomial subspaces and apply it separately to the two purification generators. This known implication is context for the failed mixed-state comparisons, rather than a new headline inequality. The early-time pure-density relation also has antecedents in Ref. [2].

Our focus is finite-dimensional, time-independent unitary dynamics. The results do not assess nonunitary examples or typical-state limits. A final fresh public-prior-art sweep before release found no matching public prior result; absolute priority is not claimed.

Section 2 fixes the definitions and spectral representation. Sections 3–5 contain the comparison results and their proofs. Section 6 gives restricted survivors and the spectral mechanism. Exact finite certificates and technical estimates are collected in the appendices.

2 Definitions and spectral measures

2.1 Seeds, generators, and normalization

Let H be a time-independent Hermitian matrix on a physical Hilbert space of dimension d , and let ρ be positive semidefinite with trace one. We set $\hbar = 1$ and use

$$U_t = e^{-itH}, \quad \rho(t) = U_t \rho U_t^\dagger, \quad P = \text{Tr } \rho^2. \quad (2.1)$$

For a Hermitian generator G and a unit vector v , orthonormalize $v, Gv, G^2v, \dots$ until exact termination. If K_n is this basis, the complexity associated with the pair (G, v) is

$$C_{G,v}(t) = \sum_{n=0}^{m-1} n |\varphi_n(t)|^2, \quad \varphi_n(t) = \langle K_n | e^{-itG} | v \rangle, \quad \sum_{n=0}^{m-1} |\varphi_n(t)|^2 = 1. \quad (2.2)$$

The integer m is the cyclic dimension, not the physical dimension. The Lanczos phase convention does not affect these probabilities.

For operator complexity the inner product is $(A|B) = \text{Tr}(A^\dagger B)$, and the generator is the self-adjoint commutator map $\mathcal{L}_H A = [H, A]$. The normalized mixed seed is $\rho/\sqrt{P}$. In particular, unit trace does not imply unit Hilbert–Schmidt norm. This normalization agrees with Ref. [3, Supplemental Eqs. (S.7)–(S.11)].

For full-rank ρ , fix row vectorization and define

$$\begin{aligned} \text{vec}_r A &= \sum_{a,b} A_{ab} |a\rangle \otimes |b\rangle, & s &= \text{vec}_r \sqrt{\rho}, & \|s\|^2 &= \text{Tr } \rho = 1, \\ G_I &= H \otimes I, & G_{U^*} &= H \otimes I - I \otimes \overline{H}, & s_X(t) &= e^{-itG_X} s. \end{aligned} \quad (2.3)$$

Thus $s_I(t) = \Psi_\rho^I(t)$ and $s_{U^*}(t) = \text{vec}_r \sqrt{\rho(t)}$. A bar denotes entrywise conjugation; for Hermitian H , $\overline{H} = H^T$. Both purifications have physical reduced density matrix $\rho(t)$. We

abbreviate

$$C_M = C_K(\rho(t)), \quad S_X = C_S(s_X(t)), \quad K_X = C_K(|s_X(t)\rangle\langle s_X(t)|), \quad X \in \{I, U^*\}. \quad (2.4)$$

The seed for K_X is $ss^\dagger$, with norm one, and its generator is $[G_X, \cdot]$. The source hierarchy is $S_{U^*} \leq C_M \leq K_I$. For rank-deficient states, statements about arbitrary purifications below use the actual normalized pure seed and its generator on a fixed enlarged space. No identification of canonical and minimal purifications is assumed.

2.2 The spectral representation

The pair (G, v) determines a probability measure ν on the eigenvalues of G . Orthonormal polynomials p_n with respect to ν represent the Krylov vectors, and

$$\varphi_n(t) = \int p_n(x) e^{-itx} d\nu(x). \quad (2.5)$$

One may choose the polynomials real. Repeated spectral values are merged; the number of distinct atoms with positive weight is the exact chain length. In an energy basis of H , the mixed and time-dependent spread measures are

$$\nu_M = \sum_{a,b} \frac{|\rho_{ab}|^2}{P} \delta_{E_a - E_b}, \quad \nu_{S,U^*} = \sum_{a,b} |(\sqrt{\rho})_{ab}|^2 \delta_{E_a - E_b}. \quad (2.6)$$

For the time-independent purified vector the energy measure is $\nu_{S,I} = \sum_a \rho_{aa} \delta_{E_a}$. Its rank-one operator measure is the difference convolution of this measure with itself:

$$\nu_{K,I} = \sum_{a,b} \rho_{aa} \rho_{bb} \delta_{E_a - E_b}. \quad (2.7)$$

More generally, a pure-state measure produces the difference convolution when its rank-one density matrix is vectorized. In contrast, the two measures in Eq. (2.6) have the same possible frequencies but different weights. Equality of frequency support does not order their complexities.

Lemma 2.1 (Three-atom chain). *For $0 < \mu < 1$, the measure $\nu_\mu = (1 - \mu)\delta_0 + (\mu/2)(\delta_{-1} + \delta_1)$ has complexity*

$$F(\mu; t) = \mu \sin^2 t + 8\mu(1 - \mu) \sin^4(t/2). \quad (2.8)$$

The same expression holds at $\mu = 0, 1$ with the corresponding terminated chains. A frequency gap ω replaces t by ωt .

Proof. The squared nonzero Lanczos coefficients are $b_1^2 = \mu$ and $b_2^2 = 1 - \mu$, and all diagonal coefficients vanish. The Jacobi matrix satisfies $J^3 = J$. Its evolved seed has amplitudes

$$(1 - \mu + \mu \cos t, -i\sqrt{\mu} \sin t, \sqrt{\mu(1 - \mu)}(\cos t - 1)). \quad (2.9)$$

Their squared moduli sum to one, and their index-weighted sum gives Eq. (2.8). At either endpoint the zero-coupling branch is simply omitted. $\square$

Lemma 2.2 (Return bound). *For a normalized finite chain of length m with return amplitude $A(t)$,*

$$1 - |A(t)|^2 \leq C(t) \leq (m - 1)(1 - |A(t)|^2). \quad (2.10)$$

For the positive-matrix flows of ρ or $\sqrt{\rho}$, $A(t)$ is real and lies in $[0, 1]$. If $m \leq 5$ and $D(t) = 1 - A(t)$, then

$$D(t) \leq C(t) \leq 8D(t). \quad (2.11)$$

Proof. The total probability away from node zero is $1 - |A|^2$, and each of its indices lies between 1 and $m - 1$. Positivity of the matrix overlap and Cauchy-Schwarz give $0 \leq A \leq 1$ in the stated flows. Finally, $1 - A^2 = (1 - A)(1 + A)$ lies between D and $2D$. $\square$

3 Exact failures of the two-sided hierarchy

3.1 The lower inequality

Proposition 3.1 (Full-rank qutrit lower-side witness). *For*

$$H_L = \text{diag}(2, 0, 1), \quad \rho_L = \frac{1}{42} \begin{pmatrix} 32 & 0 & 0 \\ 0 & 5 & 3 \\ 0 & 3 & 5 \end{pmatrix}, \quad (3.1)$$

the lower inequality in Eq. (1.1) fails at $t = \pi$:

$$S_{U^*}(\pi) = \frac{82}{441}, \quad C_M(\pi) = \frac{1074}{8281}, \quad S_{U^*}(\pi) - C_M(\pi) = \frac{4192}{74529} > 0. \quad (3.2)$$

Proof. The eigenvalues of ρ_L are $16/21, 4/21, 1/21$, all positive, and

$$\sqrt{\rho_L} = \frac{1}{2\sqrt{21}} \begin{pmatrix} 8 & 0 & 0 \\ 0 & 3 & 1 \\ 0 & 1 & 3 \end{pmatrix}, \quad P_L = \frac{13}{21}. \quad (3.3)$$

Only the second and third levels carry coherence, with energy gap one. Equation (2.6) therefore reduces both problems to Lemma 2.1, with

$$\mu_S = \frac{1}{42}, \quad \mu_K = \frac{3}{182}, \quad S_{U^*}(t) = F(\mu_S; t), \quad C_M(t) = F(\mu_K; t). \quad (3.4)$$

Evaluation at π gives Eq. (3.2) exactly. The full probability vectors appear in Appendix A. $\square$

The witness involves neither a rank-deficient limit nor a numerical truncation. The light coherent sector carries a larger relative weight in $\sqrt{\rho}$ than in $\rho/\sqrt{P}$. Appendix A.2 embeds it in a full-rank qutrit family for which $S_{U^*}(\pi)/C_M(\pi)$ is unbounded, ruling out a repair of the lower side by a fixed positive state-independent multiplier.

3.2 The upper inequality

Proposition 3.2 (Full-rank qutrit upper-side witness). *For*

$$\rho_R = \frac{1}{26} \text{diag}(16, 1, 9), \quad H_R = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad (3.5)$$

the upper inequality in Eq. (1.1) fails at $t = \pi/3$:

$$C_M(\pi/3) = \frac{1141425}{913952}, \quad K_I(\pi/3) = \frac{2967537}{2456246}, \quad C_M(\pi/3) - K_I(\pi/3) = \frac{1600683}{39299936} > 0. \quad (3.6)$$

It also fails throughout the explicit interval $0 < t < 1/2704$.

Proof. The state is strictly positive, has purity $P_R = 1/2$, and has positive square root $\text{diag}(4, 1, 3)/\sqrt{26}$. In the energy basis of H_R , the normalized mixed-operator measure and the purified-operator measure are

	ordered spectral support	ordered weights
C_M	$(-2, 0, 2)$	$(225/1352, 451/676, 225/1352)$
K_I	$(-2, -1, 0, 1, 2)$	$(289/2704, 153/676, 451/1352, 153/676, 289/2704)$

(3.7)

For the second row the purified vector has energy weights $17/52, 9/26, 17/52$ on $-1, 0, 1$; taking their difference convolution gives the stated row. All weights are positive, so the chains terminate exactly at lengths three and five. The rational probabilities obtained from Eq. (2.5) are listed in Appendix A; their weighted sums prove the finite-time claim.

The leading coefficients give a separate short-time certificate:

$$C_M(t) = \frac{225}{169}t^2 + O(t^4), \quad K_I(t) = \frac{221}{169}t^2 + O(t^4). \quad (3.8)$$

They are $\|[H_R, \rho_R]\|_{\text{HS}}^2 / P_R$ and $2 \text{Var}_{\rho_R}(H_R)$, respectively. A norm estimate for the Taylor remainder, proved in Appendix B, gives

$$C_M(t) - K_I(t) \geq \frac{4}{169}t^2 - 64t^3 > 0, \quad 0 < t < \frac{1}{2704}. \quad (3.9)$$

□

These are separate witnesses for separate inequalities. No assertion is made that one displayed state violates both sides, or that every mixed state violates either side.

3.3 Qubits and minimal physical dimension

Theorem 3.3 (Qubit hierarchy). *For every qubit density matrix, every time-independent Hermitian Hamiltonian, and every real time, $S_{U^*}(t) \leq C_M(t) \leq K_I(t)$. Consequently, physical dimension three is minimal for failure of either side of Eq. (1.1).*

Proof. Let p_1, p_2 be the density-matrix eigenvalues, and let θ be the polar rotation angle between its eigenbasis and the energy basis. Put $z = (p_1 - p_2)^2$ and $c = \cos^2 \theta$, both in $[0, 1]$. With $\tau = (E_2 - E_1)t$, the source's qubit formulas [3, Eq. (18), Supplemental Eqs. (S.24)–(S.31)] have the three-atom form

$$\begin{aligned} S_{U^*} &= F(\mu_S; \tau), & \mu_S &= \frac{(1 - \sqrt{1 - z})(1 - c)}{2}, \\ C_M &= F(\mu_K; \tau), & \mu_K &= \frac{z(1 - c)}{1 + z}, \\ K_I &= F(\mu_I; \tau), & \mu_I &= \frac{1 - zc}{2}. \end{aligned} \quad (3.10)$$

All three parameters lie in $[0, 1/2]$. The lower ordering follows, for nonzero μ_S , from

$$\frac{\mu_K}{\mu_S} = \frac{(\sqrt{p_1} + \sqrt{p_2})^2}{p_1^2 + p_2^2} \geq 1, \quad \mu_I - \mu_K = \frac{(1 - z)(1 + zc)}{2(1 + z)} \geq 0. \quad (3.11)$$

When $\mu_S = 0$, the first ordering follows directly from Eq. (3.10). The function $F(\mu; \tau)$ is nondecreasing in μ on $[0, 1/2]$, since its derivative is $\sin^2 \tau + 8(1 - 2\mu)\sin^4(\tau/2) \geq 0$. This proves the hierarchy. Pure, maximally mixed, zero-gap, and stationary endpoints are included by the displayed expressions with their actual terminated chains. In particular, stationary mixed dynamics need not imply stationary I -purification dynamics, but the ordering still holds. Dimension one is stationary, and Propositions 3.1 and 3.2 supply the dimension-three failures. □

3.4 Failure without a reducing spectator sector

Proposition 3.4 (Upper-side perturbation). *Keep ρ_R fixed and replace H_R by*

$$H_\delta = \begin{pmatrix} 0 & 1 & \delta \\ 1 & 0 & 2\delta \\ \delta & 2\delta & 0 \end{pmatrix}. \quad (3.12)$$

For $0 < |\delta| < 1/\sqrt{135}$, the pair has no nontrivial common reducing projection and the upper inequality fails at sufficiently small positive times. At $\delta = 1/20$ and $t = 1/20000$ one has the explicit certificate

$$C_M(t) - K_I(t) \geq \frac{378247}{2112500000000000} > 0. \quad (3.13)$$

Proof. The exact difference of the quadratic coefficients is

$$k_K - k_I = \frac{4 - 540\delta^2}{169} > 0. \quad (3.14)$$

Finite-dimensional analyticity proves the short-time statement. Since ρ_R has simple spectrum, a common reducing projection must select a subset of its eigenvectors. Every off-diagonal entry of H_δ is nonzero, so no proper nonempty subset reduces it. The finite-time estimate follows from the explicit norm bound in Appendix B. $\square$

4 The pure-state factor-two theorem

The two-sided mixed-state comparison and the pure-density comparison use different pairs of spectral measures. The latter has a tensor structure that protects a universal inequality.

Theorem 4.1 (Finite-dimensional factor-two inequality). *Let G be a Hermitian matrix and ψ a unit vector in a finite-dimensional Hilbert space. Set $\psi(t) = e^{-itG}\psi$ and $\sigma(t) = |\psi(t)\rangle\langle\psi(t)|$. With the state Krylov chain generated by (G, ψ) and the operator chain generated by $([G, \cdot], \sigma(0))$, for every real t ,*

$$\mathcal{C}_K(\sigma(t)) \geq 2\mathcal{C}_S(\psi(t)). \quad (4.1)$$

The coefficient two is sharp. Stationary seeds, repeated eigenvalues, and terminating chains are included.

Proof. Under row vectorization,

$$\chi = \psi \otimes \bar{\psi}, \quad L = G \otimes I - I \otimes \bar{G}, \quad \chi(t) = \psi(t) \otimes \overline{\psi(t)}. \quad (4.2)$$

Let K_i be the state Krylov basis and define the polynomial subspaces

$$V_n = \text{span}\{L^k \chi : 0 \leq k \leq n\}, \quad W_n = \text{span}\{K_i \otimes \bar{K}_j : i + j \leq n\}. \quad (4.3)$$

The binomial expansion of L^k proves $V_n \subseteq W_n$. This inclusion remains valid if a chain terminates, or if the joint cyclic subspace is a proper subspace of the tensor-product space. No completion of the joint Krylov basis is needed.

Write $p_i = |\langle K_i | \psi(t) \rangle|^2$ for the state probabilities and p_k^{op} for the operator probabilities. Conjugate evolution has the same probabilities; the possible signs $(-1)^j$ of its Krylov vectors do not affect their spans. Nested orthogonal projections give

$$\sum_{k>n} p_k^{\text{op}} = 1 - \|P_{V_n} \chi(t)\|^2 \geq 1 - \|P_{W_n} \chi(t)\|^2 = \sum_{i+j>n} p_i p_j. \quad (4.4)$$

Summing these nonnegative tails yields

$$\mathcal{C}_K(\sigma(t)) \geq \sum_{i,j} (i+j) p_i p_j = 2 \sum_i i p_i = 2\mathcal{C}_S(\psi(t)). \quad (4.5)$$

For any nonstationary finite-dimensional seed, with $v = \langle \psi | (G - \langle G \rangle)^2 | \psi \rangle > 0$, the short-time coefficients of $\mathcal{C}_K$ and $\mathcal{C}_S$ are $2v$ and v . Their ratio tends to two, so no larger universal multiplier is possible. A stationary seed gives zero on both sides. $\square$

The proof also identifies the excess complexity exactly:

$$\mathcal{C}_K(\sigma(t)) - 2\mathcal{C}_S(\psi(t)) = \sum_{n \geq 0} \|(P_{W_n} - P_{V_n})\chi(t)\|^2. \quad (4.6)$$

In finite dimension the sum terminates. Its positivity is an assertion about the evolved cyclic vector and the nested subspaces. It does not require an unqualified positivity claim for a joint number operator on an unused ambient complement.

Corollary 4.2 (The two purification branches). *For either of the source’s unitary purification schemes on a fixed finite enlarged Hilbert space, using its actual normalized initial purification,*

$$K_I(t) \geq 2S_I(t), \quad K_{U^*}(t) \geq 2S_{U^*}(t) \quad (t \in \mathbb{R}). \quad (4.7)$$

Proof. Apply Theorem 4.1 first with $G = G_I$, and separately with $G = G_{U^*}$. Both use the pure seed s and the rank-one operator seed $ss^\dagger$. The two generators need not have the same Krylov structure. The theorem itself does not require full rank of the original density matrix. $\square$

Equation (4.1) is a special case of the prior tensor-product superadditivity theorem of Murugan and van Zyl [4]: use generators G and $-\bar{G}$ and the product seed $\psi \otimes \bar{\psi}$. Their two subsystem complexities are equal. The self-contained subspace proof above fixes the source matching without making the factor-two theorem a novelty claim. We restrict the theorem here to finite dimension; no infinite-dimensional extension is required for any counterexample or no-go result in this paper.

5 Limits of temporal concentration and purity-only control

5.1 The source statement and the quantified questions

For nonstationary dynamics define, wherever the denominators are nonzero,

$$r(t) = \frac{S_{U^*}(t)}{C_M(t)}, \quad q(t) = \frac{C_M(t)}{S_{U^*}(t)} = \frac{1}{r(t)}, \quad R(t) = Pq(t). \quad (5.1)$$

The main-text Conjecture 2 of Ref. [3] concerns r . The supplement also considers the reciprocal q , and its Eq. (S.32) defines the purity-rescaled reciprocal R . These quantities must not be interchanged when formulating a concentration statement.

For a positive time series with common recurrence period T , use

$$\langle f \rangle_T = \frac{1}{T} \int_0^T f(t) dt, \quad \text{CV}(f)^2 = \frac{\langle f^2 \rangle_T}{(\langle f \rangle_T)^2} - 1. \quad (5.2)$$

The source uses the temporal standard deviation divided by the mean over a recurrence period. Its roughly five-percent observation refers to a particular Werner-state scan, not a uniform theorem. Its tabulated relative maximum fluctuation is a different quantity from the coefficient of variation. Nor does the source state an exact identity $r = f(P)$.

We consider three explicitly stronger universal formulations. First, at fixed dimension and fixed Hamiltonian, could one finite constant bound $\text{CV}(r)$ for every admissible nonstationary full-rank state? Second, could such a constant bound $\text{CV}(q)$, and hence $\text{CV}(R)$? Third, at fixed dimension, purity, and Hamiltonian, could r or its period mean have a finite upper bound determined by purity alone? A separate construction answers each question negatively. Thus the results are no-go theorems for these source-motivated universal formulations. They do not assign a truth value to every possible reading of the qualitative conjecture.

5.2 Main-ratio concentration

Theorem 5.1 (Unbounded recurrence-period variation of r). *Fix the four-dimensional Hamiltonian and two-level block*

$$H = \text{diag}(0, 2, 3, 4), \quad B = \frac{1}{10} \begin{pmatrix} 5 & 3 \\ 3 & 5 \end{pmatrix}, \quad \rho_\epsilon = (1 - \epsilon)B \oplus \epsilon B, \quad 0 < \epsilon \leq \frac{1}{4}. \quad (5.3)$$

These states are full rank. With the uniform time average over $T = 2\pi$, and with removable continuation at common recurrences,

$$\text{CV}(r)^2 \geq \frac{289}{132710400\pi\epsilon} - 1 \rightarrow \infty \quad (\epsilon \rightarrow 0^+). \quad (5.4)$$

In particular, the rational choice $\epsilon = 10^{-10}$ gives $\text{CV}(r) > 80$.

Proof. The eigenvalues are $4(1 - \epsilon)/5, (1 - \epsilon)/5, 4\epsilon/5, \epsilon/5$. Let $D_0 = (1 - \epsilon)^2 + \epsilon^2$, so $P = 17D_0/25$, and note that $\sqrt{B} = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix} / (2\sqrt{5})$. The normalized return losses are

$$\begin{aligned} D_S &= 1 - \text{Tr}(\sqrt{\rho(t)}\sqrt{\rho}) = \frac{(1 - \epsilon)(1 - \cos 2t) + \epsilon(1 - \cos t)}{10}, \\ D_K &= 1 - \frac{\text{Tr}(\rho(t)\rho)}{P} = \frac{9}{34D_0} [(1 - \epsilon)^2(1 - \cos 2t) + \epsilon^2(1 - \cos t)]. \end{aligned} \quad (5.5)$$

Both chains have exactly the five frequencies $0, \pm 1, \pm 2$ with positive weights. Lemma 2.2 yields $f_\epsilon/8 \leq r \leq 8f_\epsilon$, where, writing $c = \cos^2(t/2)$,

$$\begin{aligned} f_\epsilon &= \frac{D_S}{D_K} = \frac{17D_0}{45} \frac{4(1 - \epsilon)c + \epsilon}{4(1 - \epsilon)^2c + \epsilon^2} \\ &= \frac{17D_0}{45(1 - \epsilon)} \left[1 + \frac{\epsilon(1 - 2\epsilon)}{4(1 - \epsilon)^2c + \epsilon^2} \right]. \end{aligned} \quad (5.6)$$

The common recurrence factor has been cancelled. The elementary integral in Appendix B gives

$$\langle f_\epsilon \rangle_T = \frac{17D_0}{45(1 - \epsilon)} \left[1 + \frac{1 - 2\epsilon}{\sqrt{4(1 - \epsilon)^2 + \epsilon^2}} \right] \leq \frac{68}{81} < 1, \quad \langle r \rangle_T \leq 8. \quad (5.7)$$

On the window $|t - \pi| \leq \epsilon$, one has $c \leq \epsilon^2/4$ and $D_0 \geq 1/2$. Consequently

$$f_\epsilon \geq \frac{17}{180\epsilon}, \quad r \geq \frac{17}{1440\epsilon}, \quad \langle r^2 \rangle_T \geq \frac{289}{2073600\pi\epsilon}. \quad (5.8)$$

Combining the mean and second-moment bounds gives Eq. (5.4). At $\epsilon = 10^{-10}$, the elementary estimate $\pi < 22/7$ makes its lower bound exceed 80^2 . $\square$

At $t = \pi$ the heavily populated block has returned, while the light block has not. The light coherent sector contributes order ϵ to the $\sqrt{\rho}$ measure and order ϵ^2 to the normalized ρ measure. The resulting excursion is narrow, but its second moment is large enough to rule out any finite uniform coefficient of variation.

5.3 Reciprocal and rescaled-reciprocal concentration

Inversion does not preserve relative temporal concentration. The preceding family therefore cannot substitute for a proof about the supplement's ratio.

Theorem 5.2 (Unbounded recurrence-period variation of q). *Keep $H = \text{diag}(0, 2, 3, 4)$. Let $\eta > 0$, put $\epsilon = \eta^2$, and assume $0 < \epsilon \leq 1/16$. Define*

$$A_\eta = \eta \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \oplus \begin{pmatrix} 1 & \eta^3 \\ \eta^3 & 1 \end{pmatrix}, \quad \rho_\eta = \frac{A_\eta^2}{N}, \quad N = \text{Tr } A_\eta^2. \quad (5.9)$$

For these full-rank states, using $T = 2\pi$ and removable recurrence values,

$$\text{CV}(q)^2 \geq \frac{1}{1679616\pi\epsilon} - 1 \rightarrow \infty, \quad \text{CV}(R) = \text{CV}(q). \quad (5.10)$$

The rational choice $\eta = 1/100000$ gives $\text{CV}(q) > 40$.

Proof. The eigenvalues of A_η are $3\eta, \eta, 1 + \eta^3, 1 - \eta^3$, which are positive in the stated range. Its normalization and fourth trace are

$$N = 2 + 10\epsilon + 2\epsilon^3, \quad T_4 = \text{Tr } A_\eta^4 = 2 + 82\epsilon^2 + 12\epsilon^3 + 2\epsilon^6, \quad P = \frac{T_4}{N^2}. \quad (5.11)$$

The positive square root of ρ_η is $A_\eta/\sqrt{N}$. Write $a = 1 - \cos 2t$ and $b = 1 - \cos t$. Direct autocorrelation gives

$$D_S = \frac{2\epsilon}{N}a + \frac{2\epsilon^3}{N}b, \quad D_K = \frac{32\epsilon^2}{T_4}a + \frac{8\epsilon^3}{T_4}b. \quad (5.12)$$

Again both chains have five positive-weight atoms. Thus $g_\epsilon/8 \leq q \leq 8g_\epsilon$, with

$$g_\epsilon = \frac{D_K}{D_S} = \frac{4N\epsilon}{T_4} \frac{16c + \epsilon}{4c + \epsilon^2}, \quad c = \cos^2(t/2). \quad (5.13)$$

On the specified range, $2 \leq N, T_4 \leq 3$, and the period integral yields

$$\langle g_\epsilon \rangle_T = \frac{4N\epsilon}{T_4} \left[4 + \frac{1 - 4\epsilon}{\sqrt{4 + \epsilon^2}} \right] \leq 27\epsilon, \quad \langle q \rangle_T \leq 216\epsilon. \quad (5.14)$$

On $|t - \pi| \leq \epsilon$, where $c \leq \epsilon^2/4$,

$$g_\epsilon \geq \frac{2N}{T_4} \geq \frac{4}{3}, \quad q \geq \frac{1}{6}, \quad \langle q^2 \rangle_T \geq \frac{\epsilon}{36\pi}. \quad (5.15)$$

These inequalities prove Eq. (5.10); $\pi < 22/7$ verifies the finite rational witness. Finally P is positive and independent of time, so multiplication by P leaves the coefficient of variation unchanged. $\square$

Here the mean becomes small while finite excursions persist in a narrowing window. A decreasing absolute fluctuation is therefore compatible with unbounded fluctuation relative to the mean.

5.4 Recurrences and admissibility

In both temporal families, the return losses are positive combinations of $1 - \cos t$ and $1 - \cos 2t$. They vanish together only at $t \in 2\pi\mathbb{Z}$. For positive-matrix returns, the lower bound in Lemma 2.2 then implies that these are precisely the zero-complexity points. At each such point the two complexities have positive quadratic coefficients, so the ratio has a finite positive removable limit. The point $t = \pi$ is not a global recurrence: the gap-one block remains active for every allowed positive parameter. Each temporal integral is finite at each fixed parameter. The divergence conclusions use the displayed inequalities and do not exchange a singular state limit with the time integral.

5.5 Fixed purity

Theorem 5.3 (No finite purity-only upper control). *Fix $d = 4$, $H = \text{diag}(2, 3, 0, 1)$, and $P = 1/2$. For $0 < \epsilon < 5/18$, let*

$$a_{\pm} = \frac{1 - \epsilon \pm \sqrt{2\epsilon - 59\epsilon^2/25}}{2}, \quad \rho_{\epsilon} = \text{diag}(a_+, a_-) \oplus \epsilon B, \quad (5.16)$$

with B as in Eq. (5.3). These states are full rank and have exactly the specified purity. Their main ratio satisfies, at every real time with removable continuation at recurrences,

$$\frac{5}{18\epsilon} \left(1 - \frac{\epsilon}{10}\right) \leq r_{\epsilon}(t) \leq \frac{5}{18\epsilon}. \quad (5.17)$$

Thus r_{ϵ} and its recurrence-period mean diverge uniformly as $\epsilon \rightarrow 0^+$, whereas q_{ϵ} and Pq_{ϵ} tend uniformly to zero.

Proof. The eigenvalues are $a_+, a_-, 4\epsilon/5, \epsilon/5$. The radicand is positive, and

$$(1 - \epsilon)^2 - (2\epsilon - 59\epsilon^2/25) = (1 - 14\epsilon/5)(1 - 6\epsilon/5) > 0. \quad (5.18)$$

Therefore $a_- > 0$. Their sum is one and their squared sum is $1/2$. Only the last block carries coherence, with gap one. Its total nonzero-gap weights and exact ratio are

$$\begin{aligned} \mu_K &= \frac{9\epsilon^2}{25}, & \mu_S &= \frac{\epsilon}{10}, & x &= \sin^2(t/2), \\ r_{\epsilon}(t) &= \frac{F(\mu_S; t)}{F(\mu_K; t)} = \frac{5}{18\epsilon} \frac{1 + (1 - \epsilon/5)x}{1 + (1 - 18\epsilon^2/25)x}. \end{aligned} \quad (5.19)$$

The reduced denominator is positive. For the stated parameter interval the last fraction decreases with $x \in [0, 1]$; its endpoint estimates give Eq. (5.17). At $t \in 2\pi\mathbb{Z}$ the removable value is $5/(18\epsilon)$. The uniform bounds also apply to the period mean. $\square$

Because dimension, purity, and Hamiltonian are all fixed, the family excludes even a finite upper bound allowed to depend on those fixed data. In particular it excludes finite purity-only upper control of r or its mean and positive purity-only lower control of q or Pq . It leaves intact the weaker statement that purity influences these quantities together with additional spectral information. It does not contradict the reported behavior within the particular Werner family.

6 Restricted survivors and the spectral mechanism

6.1 A uniform qubit bound for relative variation

The failure of uniform concentration in dimension four coexists with a restricted theorem for qubits.

Proposition 6.1 (Qubit coefficient-of-variation bound). *For a nonstationary qubit under a time-independent Hermitian Hamiltonian, extend r, q, R through their removable recurrence values. For any probability-weighted time average,*

$$\text{CV}(r), \text{CV}(q), \text{CV}(R) \leq \frac{\kappa_* - 1}{2\sqrt{\kappa_*}}, \quad \kappa_* = \frac{17 + 7\sqrt{7}}{27}. \quad (6.1)$$

The upper bound is approximately 0.1375633885. No claim is made that this value is attained by a qubit trajectory and averaging measure.

Proof. Put $y = 2\sqrt{p_1 p_2} \in [0, 1)$ and $h = \sin^2 \theta > 0$; nonstationarity also requires a nonzero energy gap. The coefficients in Eq. (3.10) become

$$\mu_S = \frac{(1-y)h}{2}, \quad \mu_K = \frac{(1-y^2)h}{2-y^2}, \quad \kappa = \frac{\max_t r(t)}{\min_t r(t)} = \frac{1-\mu_S}{1-\mu_K}. \quad (6.2)$$

The same range factor applies to q and R . It increases with h , so

$$\kappa \leq \frac{(1+y)(2-y^2)}{2} \leq \frac{17+7\sqrt{7}}{27}, \quad y_{\max} = \frac{\sqrt{7}-1}{3}. \quad (6.3)$$

The second derivative of the middle expression is $-1-3y < 0$, and its endpoint values are one. For a positive random variable $X \in [m, M]$, the inequality $\mathbb{E}[(X-m)(M-X)] \geq 0$ implies

$$\text{Var}(X) \leq (M - \mathbb{E}X)(\mathbb{E}X - m), \quad \text{CV}(X) \leq \frac{M-m}{2\sqrt{mM}} = \frac{\kappa-1}{2\sqrt{\kappa}}. \quad (6.4)$$

The last step maximizes the relative variance over the allowed mean. Apply this bound to each of the three ratios. $\square$

Completely stationary 0/0 ratios are excluded from this proposition. Dimension four suffices for the temporal no-go constructions, while the qubit bound excludes dimension two. Whether dimension three suffices for unbounded temporal variation is not settled here. No minimum-dimension claim is made for the fixed-purity no-go theorem.

6.2 State-dependent comparisons

The return bound yields useful comparisons retaining the two spectral measures. Let m_S, m_K be the finite lengths of the S_{U^*} and C_M chains and define

$$A_S = \text{Tr}(\sqrt{\rho(t)}\sqrt{\rho}), \quad A_K = \frac{\text{Tr}(\rho(t)\rho)}{P}. \quad (6.5)$$

Corollary 6.2 (Return comparison). *When both return losses are nonzero,*

$$\frac{1-A_S^2}{(m_K-1)(1-A_K^2)} \leq r(t) \leq \frac{(m_S-1)(1-A_S^2)}{1-A_K^2}. \quad (6.6)$$

Proof. Divide the lower return bound for the numerator by the upper one for the denominator, and conversely. $\square$

These bounds involve return amplitudes and cyclic dimensions, rather than purity alone. Similarly, a necessary short-time condition for the source hierarchy would be

$$\|[H, \sqrt{\rho}]\|_{\text{HS}}^2 \leq \frac{\|[H, \rho]\|_{\text{HS}}^2}{P} \leq 2 \text{Var}_\rho(H), \quad \text{Var}_\rho(H) = \text{Tr}(\rho H^2) - (\text{Tr} \rho H)^2. \quad (6.7)$$

The displayed counterexamples violate the respective comparisons. Satisfying these necessary curvature inequalities is not claimed to be sufficient for an all-time hierarchy.

6.3 Why the comparisons separate

The mixed seed and the square-root seed give different powers of a population weight to a coherent sector. Hilbert–Schmidt normalization then introduces the global purity into only the former. A highly populated stationary sector can suppress mixed-state operator complexity more strongly than purification spread complexity. Independent block frequencies permit

partial recurrences that amplify this mismatch in time. Holding the total purity fixed does not fix the distribution of population among these sectors.

By comparison, a pure density matrix is a tensor product after vectorization. Its difference-convolution measure comes with the nested polynomial subspaces of Section 4. No corresponding containment is generally available for the ρ and $\sqrt{\rho}$ measures. Qubits are protected by their common three-atom form and the monotone ordering of the three coefficients. The loss of these structures, rather than an inconsistency between state and operator definitions, explains the different outcomes.

7 Discussion

Purification preserves the reduced density matrix but can change the spectral measure defining complexity. With the source's normalized seeds, each side of the mixed/purification hierarchy fails in an exact full-rank qutrit example. The qubit theorem fixes the minimum physical dimension for both failures. The upper-side perturbation further shows that its failure does not require an exactly decoupled spectator.

The ratio results delimit a different issue. Uniform temporal concentration of r and of its reciprocal fail by different four-level families, and fixed purity cannot supply finite universal upper control of r or its mean. These statements concern explicit universal formalizations of the qualitative source discussion. They neither replace that discussion by an invented quantitative conjecture nor invalidate its sample-specific numerical observations. Typical-state statements, restricted spectral families, and assumptions controlling the smallest populations would require their own quantified formulations.

The surviving factor-two inequality and qubit bounds demonstrate that useful comparisons remain possible when their spectral structure is specified. The pure-density inequality belongs to prior tensor-product superadditivity and has been included to separate that protected comparison from the failed mixed-state ordering. The state-dependent return bounds identify information that a replacement comparison can retain. No universal classification of such replacements is asserted.

All decisive constructions are finite dimensional. The temporal minimum-dimension question between three and four remains open within this analysis, and no claim is made about unrestricted infinite-dimensional or nonunitary dynamics. The accompanying exact certificates make the stated finite results directly checkable without treating numerical plots as proofs.

A Exact finite certificates

A.1 Complete probabilities for the two witnesses

For the lower-side witness at $t = \pi$, the probability vectors are

$$p^M = \left(\frac{7744}{8281}, 0, \frac{537}{8281} \right), \quad p^{S,U^*} = \left(\frac{400}{441}, 0, \frac{41}{441} \right). \quad (\text{A.1})$$

Each has unit sum. Multiplying the final entries by two gives the complexities in Eq. (3.2).

For the upper-side witness, the squared nonzero Lanczos coefficients are

$$(b_n^2)_M = \left(\frac{225}{169}, \frac{451}{169} \right), \quad (b_n^2)_{K,I} = \left(\frac{17}{13}, \frac{43}{26}, \frac{1377}{1118}, \frac{451}{559} \right). \quad (\text{A.2})$$

All diagonal coefficients vanish. At $t = \pi/3$ the full probabilities are

$$\begin{aligned} p^M &= \left(\frac{458329}{1827904}, \frac{675}{2704}, \frac{913275}{1827904} \right), \\ p^{K,I} &= \left(\frac{1500625}{7311616}, \frac{62475}{140608}, \frac{45778977}{157199744}, \frac{7803}{140608}, \frac{1173051}{314399488} \right). \end{aligned} \quad (\text{A.3})$$

For a direct finite check, each row sums to one and its index-weighted sum is the corresponding rational number in Eq. (3.6). These data follow from the exact measures in Eq. (3.7). The terminal polynomials are $x(x-2)(x+2)$ and $x(x-2)(x-1)(x+1)(x+2)$, respectively, so no floating-point stopping criterion is involved. The authoritative certificate additionally compares these spectral calculations with direct matrix-commutator Lanczos calculations of every probability.

A.2 An unbounded qutrit ratio family

Proposition A.1 (No fixed positive multiplier for the lower side). *For every real $m \geq 4$, keep $H_L = \text{diag}(2, 0, 1)$ and take*

$$\rho_m = \frac{1}{m^2 + 5} \begin{pmatrix} m^2 & 0 & 0 \\ 0 & 5/2 & 3/2 \\ 0 & 3/2 & 5/2 \end{pmatrix}. \quad (\text{A.4})$$

The state is full rank, and $S_{U^}(t) > C_M(t)$ for every $t \notin 2\pi\mathbb{Z}$. Moreover,*

$$\frac{S_{U^*}(\pi)}{C_M(\pi)} \sim \frac{m^2}{9} \rightarrow \infty \quad (m \rightarrow \infty). \quad (\text{A.5})$$

Thus no constant $c > 0$ can make $cS_{U^} \leq C_M$ hold for all full-rank qutrit states, Hermitian Hamiltonians, and real times.*

Proof. The spectrum is $(m^2, 4, 1)/(m^2 + 5)$ and

$$P_m = \frac{m^4 + 17}{(m^2 + 5)^2}, \quad \mu_S = \frac{1}{2(m^2 + 5)}, \quad \mu_K = \frac{9}{2(m^4 + 17)}. \quad (\text{A.6})$$

Their difference has numerator $m^4 - 9m^2 - 28$, which is positive for $m \geq 4$. Both complexities are given by F , and

$$F(a; t) - F(b; t) = (a - b) [\sin^2 t + 8(1 - a - b) \sin^4(t/2)]. \quad (\text{A.7})$$

Here $a = \mu_S > b = \mu_K$ and $a + b < 1$, proving the all-time strict inequality away from recurrences. At π , $F(\mu; \pi) = 8\mu(1 - \mu)$, which gives Eq. (A.5). Every finite m is an admissible full-rank state; the limiting rank-deficient state is not used as a witness. $\square$

A.3 A rational fixed-purity anchor

At $\epsilon = 1/10$, the family in Theorem 5.3 is

$$\rho_{1/10} = \frac{1}{100} \begin{pmatrix} 66 & 0 & 0 & 0 \\ 0 & 24 & 0 & 0 \\ 0 & 0 & 5 & 3 \\ 0 & 0 & 3 & 5 \end{pmatrix}, \quad P = \frac{1}{2}. \quad (\text{A.8})$$

This supplies a finite rational comparison state at the same dimension, purity, and Hamiltonian as the entire family. It prevents confusing the fixed-purity conclusion with an effect obtained solely by changing purity.

B Technical bounds

B.1 Controlled finite-time Taylor remainders

For a complexity written in its finite cyclic space as $C(t) = \langle v | e^{itL} N e^{-itL} | v \rangle$, with $N|v\rangle = 0$, repeated commutators give

$$|C'''(t)| \leq (2\|L\|)^3 \|N\|, \quad |C(t) - kt^2| \leq \frac{(2\|L\|)^3 \|N\|}{6} |t|^3, \quad (\text{B.1})$$

where k is the quadratic coefficient. For Eq. (3.5), both Liouvillian norms are at most two and the number-operator norms are two and four. Adding the two remainder estimates yields $64|t|^3$, proving Eq. (3.9).

For H_δ at $\delta = 1/20$, the maximal absolute row sum bounds $\|H_\delta\| \leq 11/10$. Each operator chain has at most seven distinct energy-gap atoms, so each number-operator norm is at most six. The two Liouvillian norms are at most $11/5$. Hence

$$C_M(t) - K_I(t) \geq \frac{53}{3380} t^2 - \frac{21296}{125} |t|^3. \quad (\text{B.2})$$

At $t = 1/20000$ this is exactly the positive rational lower bound in Eq. (3.13). These are conservative finite-time bounds, not numerical estimates of the exact differences.

B.2 Period integral and mean estimates

For $a, b > 0$, tangent substitution after splitting the interval at π gives

$$\frac{1}{2\pi} \int_0^{2\pi} \frac{dt}{a \cos^2(t/2) + b} = \frac{1}{\sqrt{b(a+b)}}. \quad (\text{B.3})$$

For the main-ratio family, $D_0 \leq 1$, $1 - \epsilon \geq 3/4$, and $\sqrt{4(1 - \epsilon)^2 + \epsilon^2} \geq 2(1 - \epsilon)$ bound the mean in Eq. (5.7) by $68/81$. On the peak window, its rational denominator is at most $2\epsilon^2$ while its numerator is at least ϵ ; together with $D_0 \geq 1/2$ this proves Eq. (5.8).

For the reciprocal family, the positive coefficients in Eq. (5.11) show that $2 \leq N, T_4 \leq 3$ on $0 < \epsilon \leq 1/16$. The bracket in Eq. (5.14) is at most $9/2$, so its prefactor bounds the mean by 27ϵ . On the window the denominator $4c + \epsilon^2$ is at most $2\epsilon^2$ and the numerator $16c + \epsilon$ is at least ϵ . This gives $g_\epsilon \geq 2N/T_4 \geq 4/3$ and hence the second-moment estimate.

B.3 The fixed-purity endpoint bound

In Eq. (5.19) the fractional-linear factor decreases because $\epsilon/5 > 18\epsilon^2/25$ for $0 < \epsilon < 5/18$. Its value at $x = 0$ is one. At $x = 1$ it is

$$\frac{1 - \epsilon/10}{1 - 9\epsilon^2/25} \geq 1 - \epsilon/10. \quad (\text{B.4})$$

This gives the lower bound in Eq. (5.17).

C Certificate map and reproducibility

The preserved internal manuscript archive contains the complete supplied input bundle, integrity records, and certification materials. The public repository provides a curated reproducibility view and does not redistribute all internal archival or third-party materials. The scientific authority is the archive `Krylov_PRL_WORK_FREEZE_2026-09-23_v3.zip`. Its SHA-256 is recorded in the project's integrity report and checked against the externally specified

anchor before use. The nested second-run archive and first-run certificate retain their original bytes.

For a compact map, the first-run certificate is the V3 directory `prior_left/P2_3_Krylov_certificate/`; the second-run certificate directory is `current_final/Krylov_remaining_attack/certificates/`. The former supplies the lower-side qutrit and its unbounded family. Within the latter, `RIGHT_BOUND_ONE_PAGE.md` and `RIGHT_BOUND_DETAILS.md` supply the upper-side witness, its perturbation, and minimality. `FACTOR_TWO_THEOREM.md` supplies the polynomial-subspace proof. `TARGET2_TEMPORAL_NOGO.md` and `TARGET2_PURITY_NOGO.md` supply the ratio families and all constants retained here. `RESTRICTED_SURVIVORS.md` supplies the qubit variation and return bounds. Exact probability data are retained in the first-run results and the second-run `results/target1r_exact.json`. The accompanying `CLAIM_SOURCE_MAP.md` gives complete paths and the scope of each theorem and headline statement.

The supplied certification evidence includes two adversarial AI audits and two frozen AI clean-room reproductions of seven specified core targets. Both clean-room runs reproduce the two finite qutrit certificates, the qubit hierarchy, the finite-dimensional factor-two result, and the conclusions of the three ratio no-go constructions. Their task was to reproduce supplied witnesses and families, not independently discover those inputs. This is AI certification evidence; it is not independent human validation or peer review. Secondary byproducts are not silently included in the clean-room coverage.

Different clean-room estimates of concentration constants are recorded only in `PROPOSED_NONAUTHORITATIVE_IMPROVEMENTS.md`. All theorem constants in this manuscript remain those of V3. A stronger estimate would require a separate targeted audit and a new scientific freeze before promotion into the claim set.

The adversarial audits identified a stale-manifest defect in the expanded tree of the historical V2 packaging. V3 records the packaging repair and preserves the unchanged authoritative mathematical archives. Replays must use a disposable extraction: the original combined verifier writes a results file and therefore changes a frozen tree if run in place. The manuscript project keeps its input archives immutable. The certification package described here did not itself perform any public-release, submission, or author-contact action.

AI contribution statement

AI systems played the primary and largest role in the scientific work reported here. They identified the source paper for further investigation, developed exact analyses of its conjectures, constructed the counterexamples, derived the proofs and auxiliary results, produced and checked the computational certificates, and assisted in the adversarial audits, clean-room reproductions, and preparation of the manuscript. Dehao Lin designed and supervised the overall research workflow, made the project and release decisions, reviewed the resulting artifacts, and assumes responsibility for the final manuscript.

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